

Week 1 Assignment – E-commerce EDA Scenario Challenge

Phase 1 – Data Audit

The dataset contains 500,000 records and 30 columns. It consists of numerical, categorical, binary, and date features. Missing values are present in columns such as `return_reason`, `shipping_cost`, `days_to_deliver`, `customer_age`, `customer_lifetime_value`, `rating`, `pages_viewed`, and `session_duration_mins`. Duplicate rows should be checked before modeling. The `order_date` column should be converted to datetime format during preprocessing.

Phase 2 – Pricing Distribution

The pricing-related variables (`product_base_price`, `final_price`, and `total_amount`) exhibit positive skewness. Histograms and skewness statistics indicate that a majority of orders are low to moderately priced, while a small number of high-value orders create a long right tail in the distribution.

Phase 3 – Categorical Columns

The dataset contains several categorical features including `category`, `sub_category`, `shipping_method`, `payment_method`, `customer_gender`, `country`, `city`, and `loyalty_tier`. High-cardinality columns such as `city` should be handled carefully, while `loyalty_tier` is an ordinal feature that should be encoded in the order Bronze < Silver < Gold < Platinum.

Phase 4 – Product Returns

The target variable `is_returned` can be analyzed to identify return patterns. Potential factors influencing returns include customer ratings, delivery time, discounts, loyalty tier, and product category. The `return_reason` column provides business insights only for returned orders and is expected to contain missing values for non-returned products.

Phase 5 – Feature Engineering

A preprocessing pipeline can be built using `ColumnTransformer`. Numerical features should be imputed with the median and scaled using `StandardScaler`. Highly skewed numerical features such as `total_amount` can be processed using `RobustScaler`. Nominal categorical variables should be one-hot encoded, while `loyalty_tier` should be ordinal encoded.

Phase 6 – Correlation Analysis

Correlation analysis should be performed on numerical variables to identify multicollinearity and relationships with the target variable `is_returned`. Features such as `days_to_deliver`, `rating`, `discount_percentage`, `customer_lifetime_value`, and purchase behavior metrics may provide useful predictive information for future return prediction models.